

YOLO Model Testing Report

Model:	yolov10x
Model Size:	61.4 MB
Parameters:	31.8 M
FLOPs (640x640):	171.85 GFLOPs
Run Name:	yolov10x_testing_20251123_234252
W&B Run Name:	yolov10x_bdd100k_yolo_test_20251123_234252
Timestamp:	2025-11-23 23:57:23
Dataset:	bdd100k_yolo - test split
Images Processed:	20000
Total Execution Time	847.65s
IoU Threshold:	0.5

Inference Performance

Metric	Value
Average Inference Time	39.10 ms per image
FPS (Frames Per Second)	25.57

Note: Inference time includes preprocessing, model inference, and postprocessing.

Overall Accuracy Metrics

Metric	Value
Precision	0.3110
Recall	0.6293
F1-Score	0.4163
mAP@0.5	0.0997

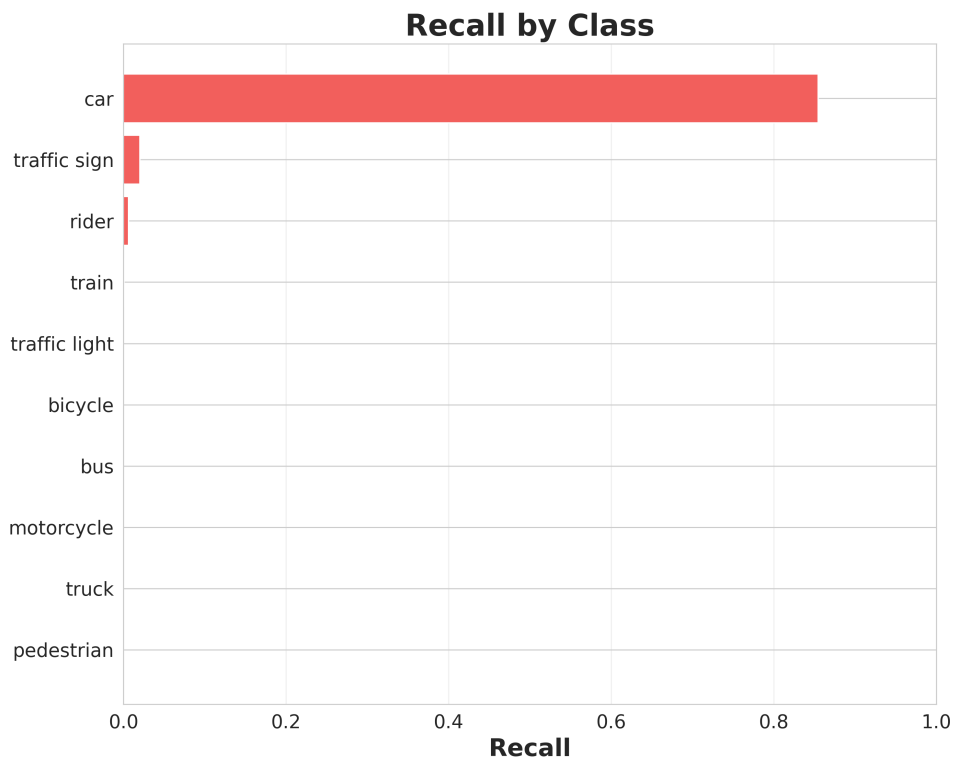
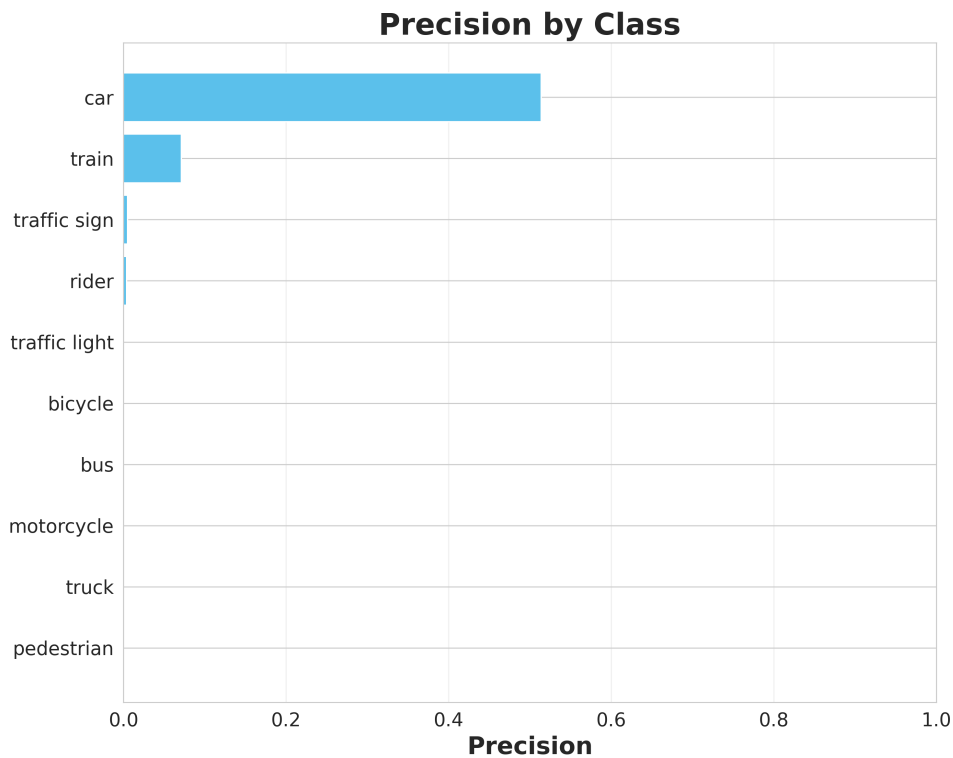
mAP@0.5:0.95	0.0601
True Positives	105793
False Positives	234348
False Negatives	62317

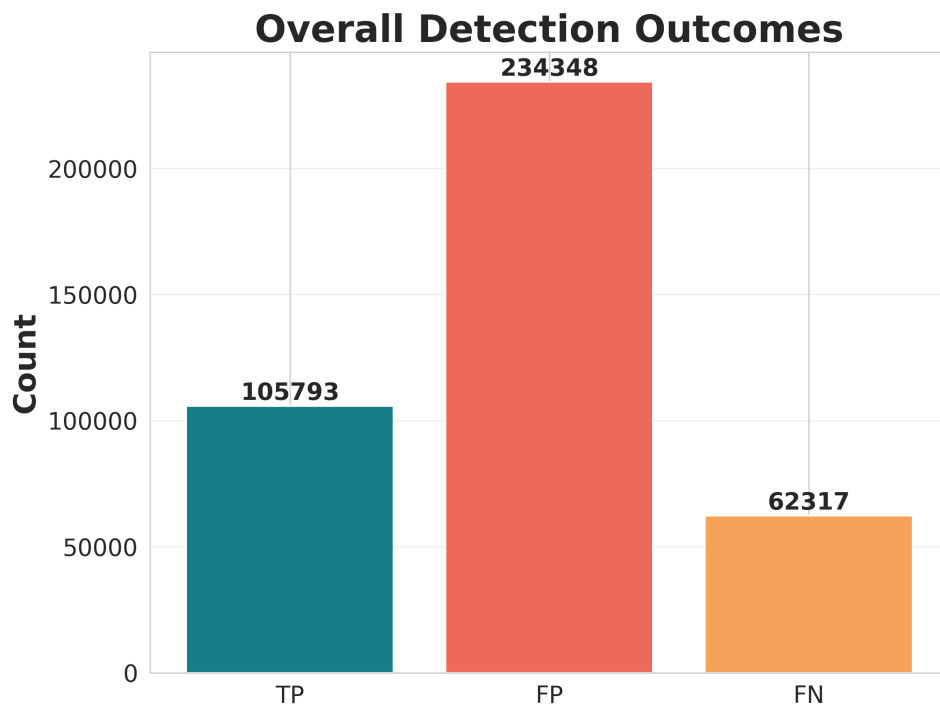
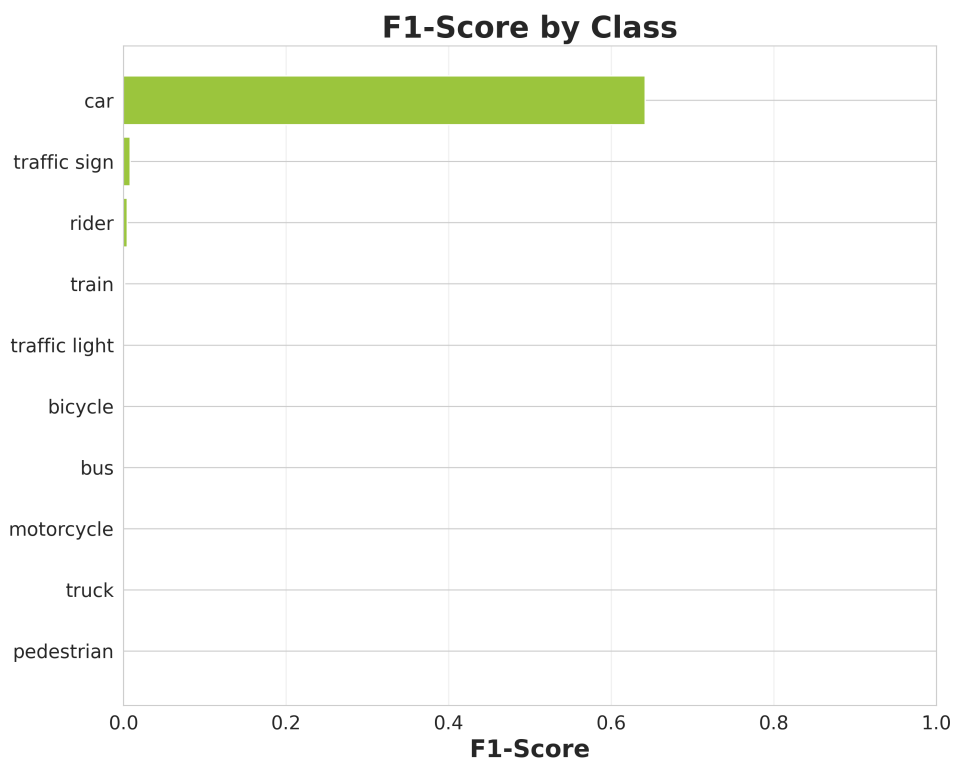
Confusion Matrix

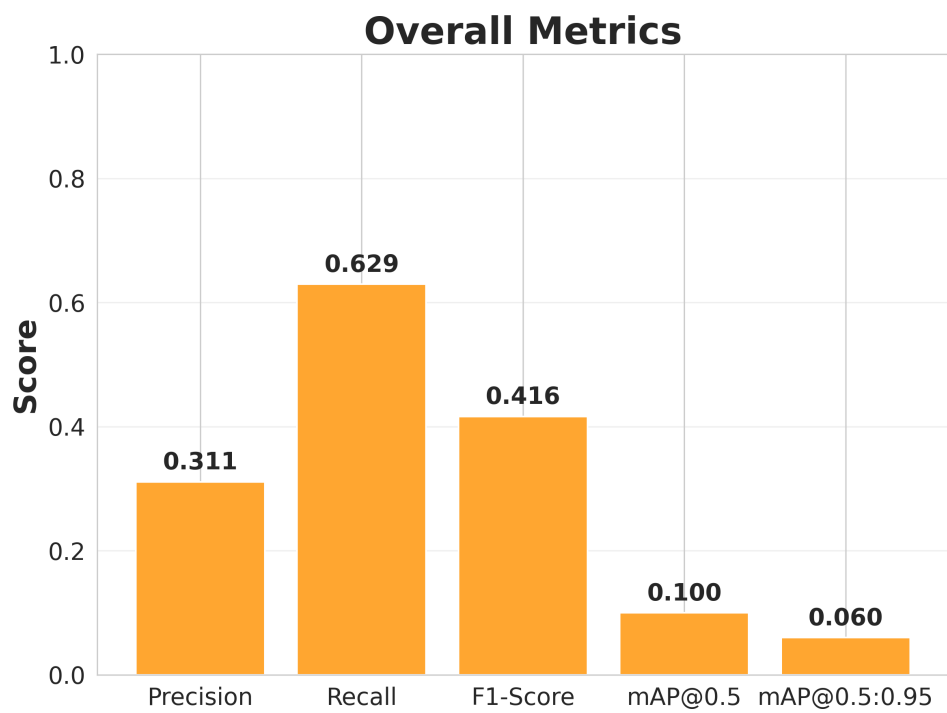
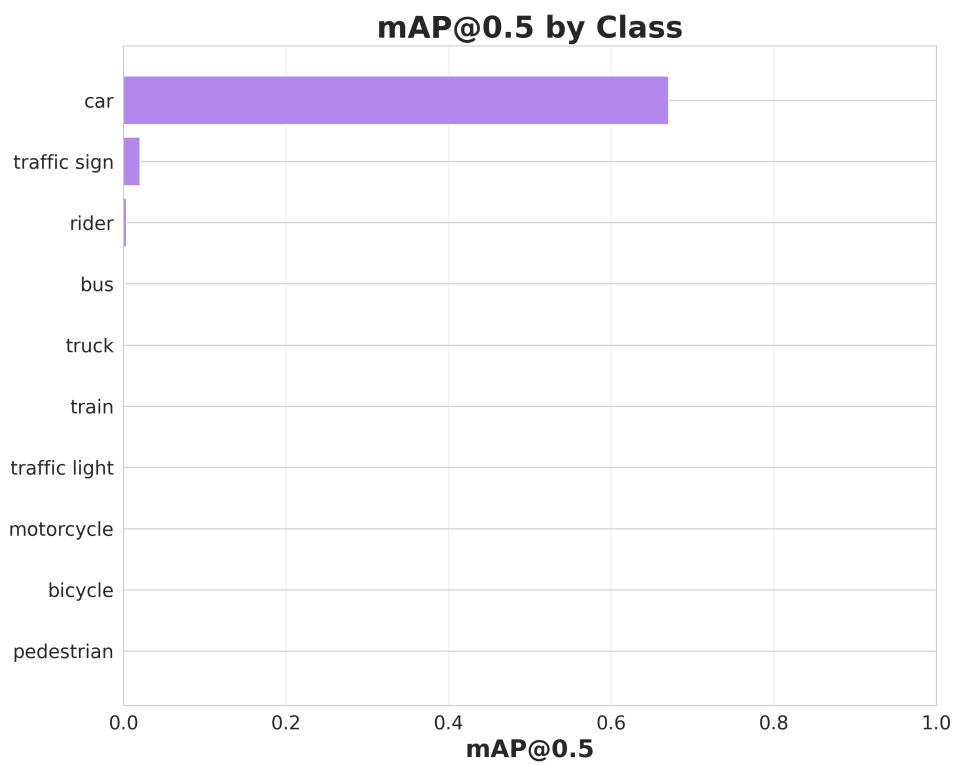
Confusion Matrix (yolov10x validation)											
True Class	pedestrian	664	72	1					3	11	
	rider	5	2			1					
	car	9	105424	856	185				20	63	
	truck	10	8								
	bus										
	train		316	293	1359	2				1	
	motorcycle			7	8	8					
	bicycle		3490	4050	375	2			2	12	
	traffic light		1	1							
	traffic sign		5	2		3			12194	362	
		pedestrian	rider	car	truck	bus	train	motorcycle	bicycle	traffic light	traffic sign
Predicted Class											

Correct Predictions (Diagonal Sum): 105793
Total Matched Predictions: 380233

Performance Visualizations







Per-Class Performance

Class	TP	FP	FN	Precision	Recall	F1-Score	mAP@0.5
pedestrian	0	0	13393	0.0000	0.0000	0.0000	0.0000
rider	5	1288	780	0.0039	0.0064	0.0048	0.0040
car	105424	99669	17922	0.5140	0.8547	0.6420	0.6712
truck	0	8701	445	0.0000	0.0000	0.0000	0.0004
bus	0	3217	10	0.0000	0.0000	0.0000	0.0012
train	2	26	2348	0.0714	0.0009	0.0017	0.0003
motorcycle	0	0	85	0.0000	0.0000	0.0000	0.0000
bicycle	0	0	9998	0.0000	0.0000	0.0000	0.0000
traffic light	0	52817	10	0.0000	0.0000	0.0000	0.0000
traffic sign	362	68630	17326	0.0052	0.0205	0.0084	0.0209

Sample Predictions: Ground Truth vs Model

Sample #1: f1c9d7ae-484fba3f.jpg

Ground Truth Objects: 15 | Predicted Objects: 8

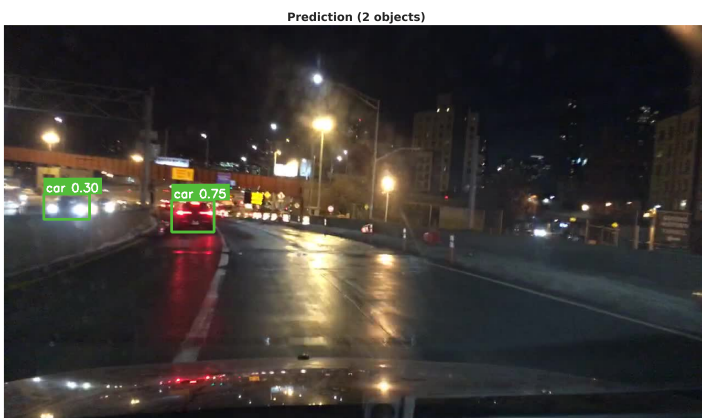
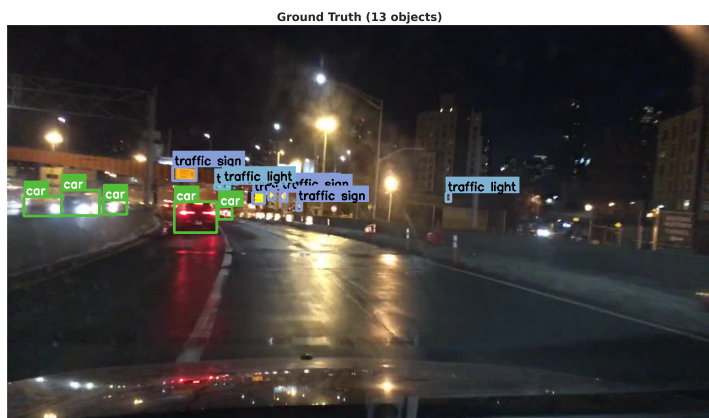
Weather: clear | Scene: highway | Time of Day: daytime



Sample #2: d6f84977-b369cf4d.jpg

Ground Truth Objects: 13 | Predicted Objects: 2

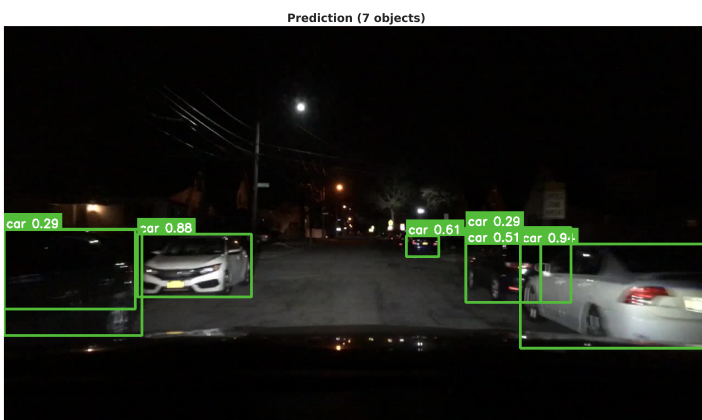
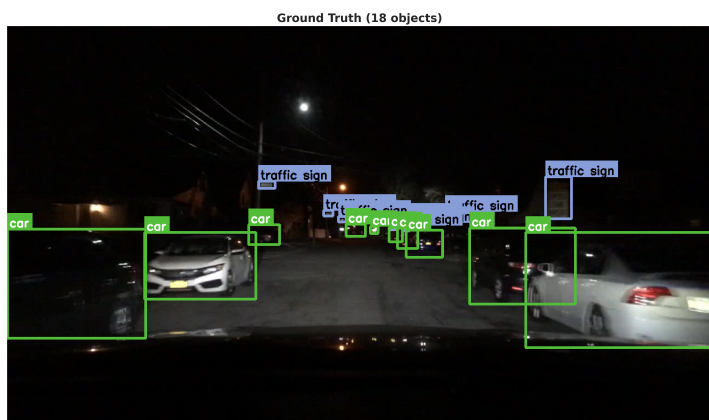
Weather: clear | Scene: highway | Time of Day: night



Sample #3: cc3bb4ea-ae675052.jpg

Ground Truth Objects: 18 | Predicted Objects: 7

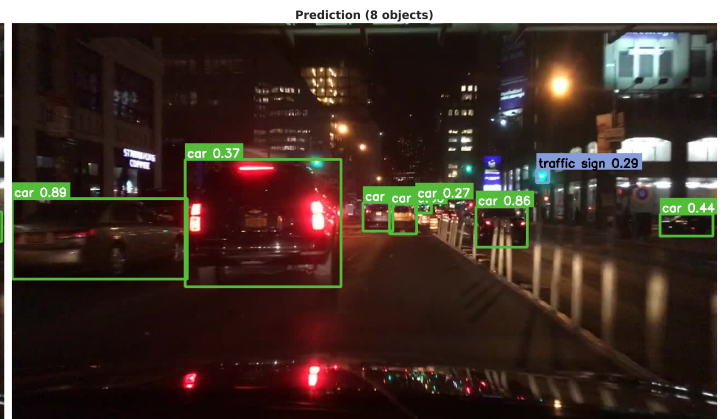
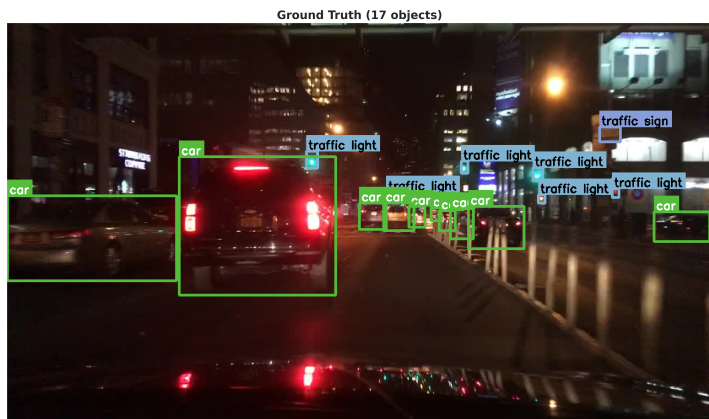
Weather: clear | Scene: city street | Time of Day: night



Sample #4: e226dce6-c4062539.jpg

Ground Truth Objects: 17 | Predicted Objects: 8

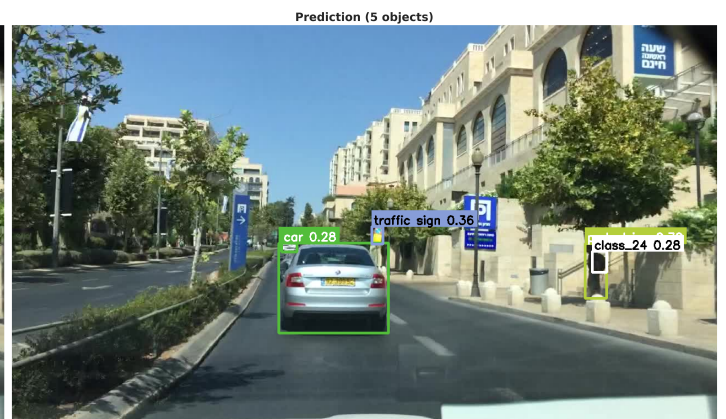
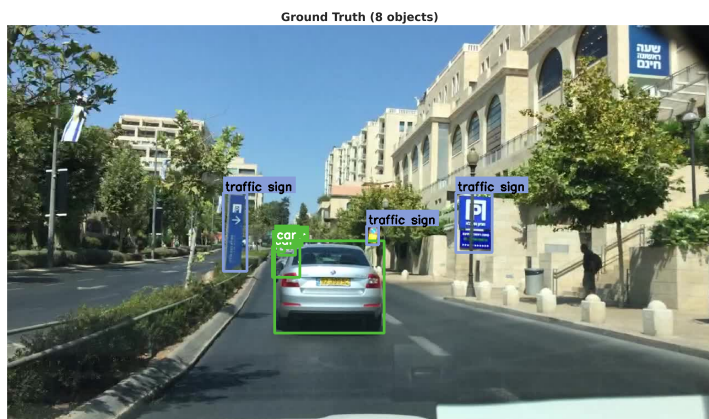
Weather: clear | Scene: city street | Time of Day: night



Sample #5: d5189b56-ae665091.jpg

Ground Truth Objects: 8 | Predicted Objects: 5

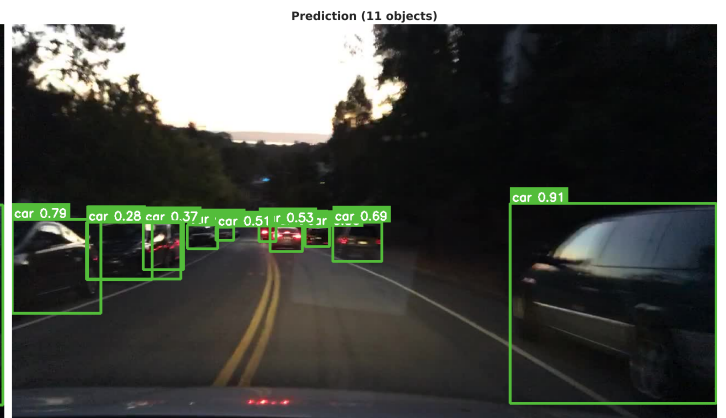
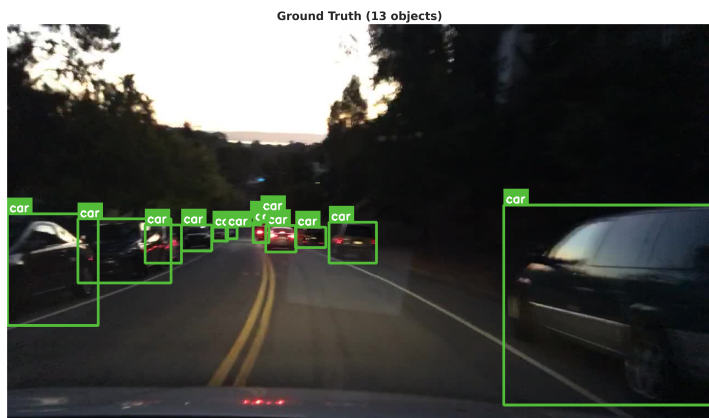
Weather: clear | Scene: residential | Time of Day: daytime



Sample #6: e0fef0d1-754a168e.jpg

Ground Truth Objects: 13 | Predicted Objects: 11

Weather: clear | Scene: highway | Time of Day: dawn/dusk



Generated by YOLO Quick Test Script